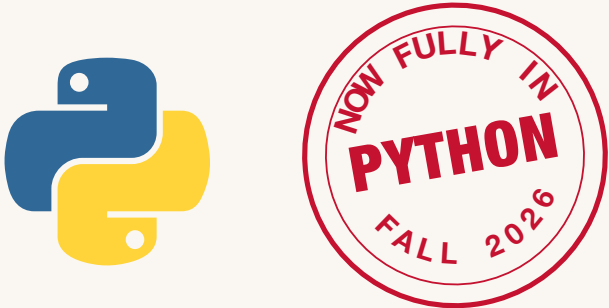


15-322 / 15-622 | FALL 2026

INTRODUCTION TO COMPUTER MUSIC

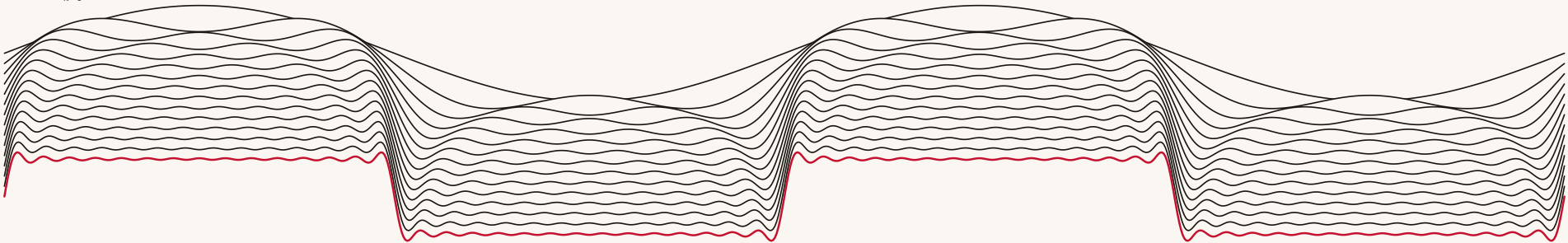
Turn code into sound.

Sampling, synthesis, Fourier analysis, effects, and generative music,
from first principles to real-time systems.



$$x_N(t) = \frac{4}{\pi} \sum_{n=1}^N \frac{\sin(2\pi(2n-1)f_0t)}{2n-1}$$

SQUARE WAVE PARTIAL SUMS · N = 1 TO 15



TUE + THU
11:00 AM – 12:20 PM



CIC 1203



9 UNITS · 15-322
12 UNITS · 15-622

WHAT YOU’LL LEARN

01

DIGITAL AUDIO

SAMPLING · QUANTIZATION · ALIASING

— signal • 16 samples — quantized

02

SYNTHESIS

ADDITIVE · FM · PHYSICAL MODELING

$$x(t) = \sum_{k=1}^3 a_k \sin(2\pi k f_0 t)$$

FM $y(t) = \sin(2\pi f_c t + I \sin(2\pi f_m t))$

03

FOURIER & DSP

DFT / FFT · FILTERS · CONVOLUTION · STFT

33-TAP WINDOWED-SINC LOW-PASS

04

EFFECTS & SPATIALIZATION

DELAY LINES · MODULATION · PHASE VOCODING

— $h_L[n]$ — $h_R[n]$

05

ALGORITHMIC COMPOSITION

MARKOV CHAINS · LPC · MIDI

P(NEXT CURRENT)	C4	E4	G4
C4	.10	.60	.30
E4	.30	.20	.50
G4	.60	.30	.10

rows sum to 1 · each note draws the next

06

REAL-TIME & MUSIC AI

STREAMING · MIR · GENERATIVE MODELS

N=512 / HOP=256 / 50% OVERLAP

audio in → buffer → |STFT| → features → model → control

one prediction per hop · 256 / 44100 s = 5.8 ms



OPEN-ENDED PROJECTS
EXPLORE YOUR OWN INTERESTS



PREREQUISITE
15-122 OR 15-112



MUSIC EXPERIENCE
NOT REQUIRED